(i)(a) Since Both sides are non-negative, squaring preserves the inequality

$$4x^2 - 8x + 64 \leq x^2 + 16x + 64$$

$$3x^2 - 24x \leq 0 \implies 3x(x-8) \leq 0$$

$$\boxed{0 \leq x \leq 8}$$

(i)(b)

$$4x^2 - 8x + 64 \leq 9x^2 - 48x + 64$$

$$5x^2 - 40x \geq 0 \implies 5x(x-8) \geq 0$$

$$\boxed{x \leq 0 \text{ or } x \geq 8}$$

(ii)(a)

$$\begin{aligned}\left(\sqrt{4x^2-8x+64} + 2(x-1)\right) \mathrm{f}(x) &= \left(4x^2-8x+64\right) - 4(x-1)^2 \\ &= \left(4x^2-8x+64\right) - \left(4x^2-8x+4\right) \\ &= 60\end{aligned}$$

$$\mathrm{f}(x) = \frac{60}{\sqrt{4x^2-8x+64} + 2(x-1)}$$

As $x \to \infty$, the denominator $\to \infty$

$$\implies \boxed{\mathrm{f}(x) \to 0}$$

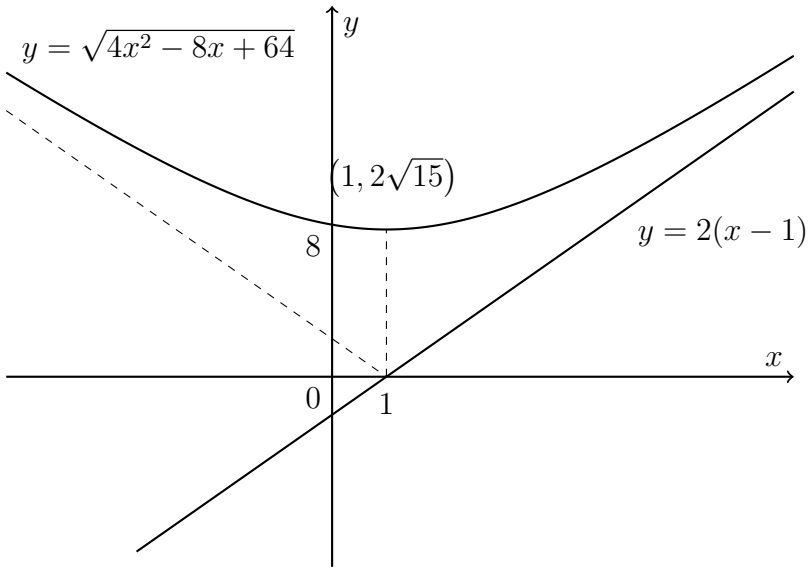
(ii)(b)

$$y = \sqrt{4x^2-8x+64} = 2\sqrt{(x-1)^2+15}$$

$$\frac{y^2}{60} - \frac{(x-1)^2}{15} = 1, \qquad y > 0$$

Minimum point $\left(1, 2\sqrt{15}\right)$, y-intercept 8, asymptotes $y = \pm 2(x-1)$

$$\mathrm{f}(x) > 0 \implies \sqrt{4x^2-8x+64} > 2(x-1) \text{ for all } x$$



(iii)

$$4x^2 - 5x + 4 = 4\left(x - \frac{5}{8}\right)^2 + \frac{39}{16} > 0$$

$$4x^2 - 5x + 4 \leq m^2x^2 + 2mcx + c^2$$

$$\left(m^2 - 4\right)x^2 + (2mc + 5)x + c^2 - 4 \geq 0$$

The solution set $\{x : x \geq 3\}$ is a half-line, so the quadratic term must vanish.

$$m^2 - 4 = 0 \implies m = \pm 2$$

Take $m = 2$:

$$(4c + 5)x + c^2 - 4 \geq 0$$

$$4c + 5 > 0 \quad \text{and} \quad c^2 - 4 = -3(4c + 5)$$

$$c^2 + 12c + 11 = 0 \implies (c + 1)(c + 11) = 0 \implies c = -1,\ -11$$

$$c = -11 \implies 4c + 5 = -39 < 0 \text{ (rejected)}$$

$$\boxed{m = 2, \quad c = -1}$$

Check:

$$4x^2 - 5x + 4 \leq (2x - 1)^2 = 4x^2 - 4x + 1 \implies x \geq 3$$

(iv)

$$\left|x^2+px+q\right| \le mx+c \iff -(mx+c) \le x^2+px+q \le mx+c$$

$$x^2+(p-m)x+(q-c) \le 0 \qquad \text{and} \qquad x^2+(p+m)x+(q+c) \ge 0$$

The first gives a closed interval $[\alpha, \beta]$, the second gives everything outside an open interval (γ, δ).

$$\{-5 \le x \le 1\} \cup \{5 \le x \le 7\} = [-5,7] \setminus (1,5)$$

$$\implies \alpha = -5,\ \beta = 7,\ \gamma = 1,\ \delta = 5$$

$$x^2+(p-m)x+(q-c) = (x+5)(x-7) = x^2-2x-35$$

$$x^2+(p+m)x+(q+c) = (x-1)(x-5) = x^2-6x+5$$

$$\begin{cases} p-m=-2 \\ p+m=-6 \end{cases} \implies \begin{cases} p=-4 \\ m=-2 \end{cases}$$

$$\begin{cases} q-c=-35 \\ q+c=-5 \end{cases} \implies \begin{cases} q=-15 \\ c=-20 \end{cases}$$

Since $\left|x^2+px+q\right| \ge 0,\ mx+c = 20-2x \ge 0$ on $-5 \le x \le 7$

This solution was submitted by **Jon**

Published and refined by **Jon**

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